Sai Krishna Rao Anugu

AI Software Engineer | Data Scientist

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Experience

AI Software Engineer 10/2025 – Present
A-Team Event GmbH *Siegen, Germany*

- Contributing to building a new startup by taking responsibility for both frontend and backend development.
- Designing and maintaining the company website and frontend components for internal and external users.
- Developing and managing backend systems and relational databases using MySQL and PostgreSQL.
- Built Python scripts for data preparation and processing, including text preprocessing using NLP techniques and LLM-based workflows.
- Supported project coordination and technical decision-making while working closely with the founding team.

Data Scientist - Master Thesis 01-06/2024
Robert Bosch GmbH – eBike systems *Kusterdingen, Baden-Württemberg, Germany*

- Topic: Comprehensive Database Development and Anomaly Detection in eBike drive units using Generative AI.
- Designed and implemented anomaly detection pipelines for spectrogram images using deep learning models.
- Developed and orchestrated ETL pipelines with Airflow.
- Built structured SQLite databases integrated with Python-based modeling.
- Deployed trained models on Databricks and cloud environments (AWS).
- Collaborated with domain experts to translate requirements into AI workflows.

Data Scientist and Machine Learning Engineer (Intern) 07-12/2023
Robert Bosch GmbH – eBike systems *Kusterdingen, Baden Württemberg, Germany*

- Designed and automated pipelines to preprocess large volumes of acoustic lab data.
- Developed automated data analysis pipelines, improving efficiency by 30%.
- Built KPI tracking models and dashboards using Power BI and Tableau.
- Implemented automated reporting pipelines using Python and SQL.
- Worked in Agile Scrum-based workflows.

Data Analyst Trainee (Remote) 01.2023 – 02.2023
Brainnest *Bremen, Germany*

- Developed data analysis logic with Python and SQL.
- Managed relational databases for KPI reporting.
- Created Tableau dashboards for performance tracking.

Working Student, Machine Learning 06.2022 – 01.2023
Intelligent Systems Group (ISG), Universität Siegen *Siegen, NRW, Germany*

- Built recommender system models using MovieLens dataset.
- Applied statistical analysis to evaluate model robustness.

Graduate Engineer Trainee (GET) 06.2019 – 06.2020
Network Enhancers India Pvt. Ltd. *Hyderabad, Telangana, India*

- Collaborated with teams to deploy and maintain network solutions.
- Resolved technical issues to reduce downtime.

Projects

SmartDoc AI: Local RAG-based Knowledge Assistant 10.2024

Siegen, Germany

- Developed an end-to-end Retrieval-Augmented Generation (RAG) system using LangChain and Mistral LLM via Ollama for fully local inference.
- Implemented semantic document retrieval using FAISS vector database with HuggingFace sentence-transformer embeddings.
- Designed and exposed RESTful API endpoints to handle document ingestion, query processing, and response generation for modular backend integration.
- Built a modern Streamlit-based interactive UI with session lifecycle management and automatic local data cleanup.
- Architected the system for privacy-sensitive document processing, ensuring no external API calls or cloud data transmission.

Education

Universität Siegen Siegen, North Rhine-Westphalia, Germany

Masters, Mechatronics with Specialization in AI *2021 - 2024*

Jawaharlal Nehru Technological University Hyderabad, Telangana, India

Bachelor of Technology, Mechanical Engineering *2015 - 2019*

Technical Skills

Programming & Databases: Python, R, SQL, C/C++, PostgreSQL, MySQL, SQLite, NoSQL
Machine Learning & AI: Supervised/Unsupervised Learning, Deep Learning, Generative AI, NLP, LLMs
Frameworks & Libraries: NumPy, Pandas, TensorFlow, PyTorch, Scikit-learn, OpenCV
Data Visualization & Analytics: Power BI, Tableau, Looker Studio, SAC
Cloud & Platforms: Databricks, AWS SageMaker, Azure, Google Cloud, Snowflake
Data Engineering & ETL: Data Collection, Cleaning, Wrangling, Validation, Airflow Pipelines
DevOps & Agile: Git, GitHub, CI/CD, Docker, Kubernetes, Scrum

Languages

- English (Fluent)
- German (B1)